```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import Image
%matplotlib inline
```

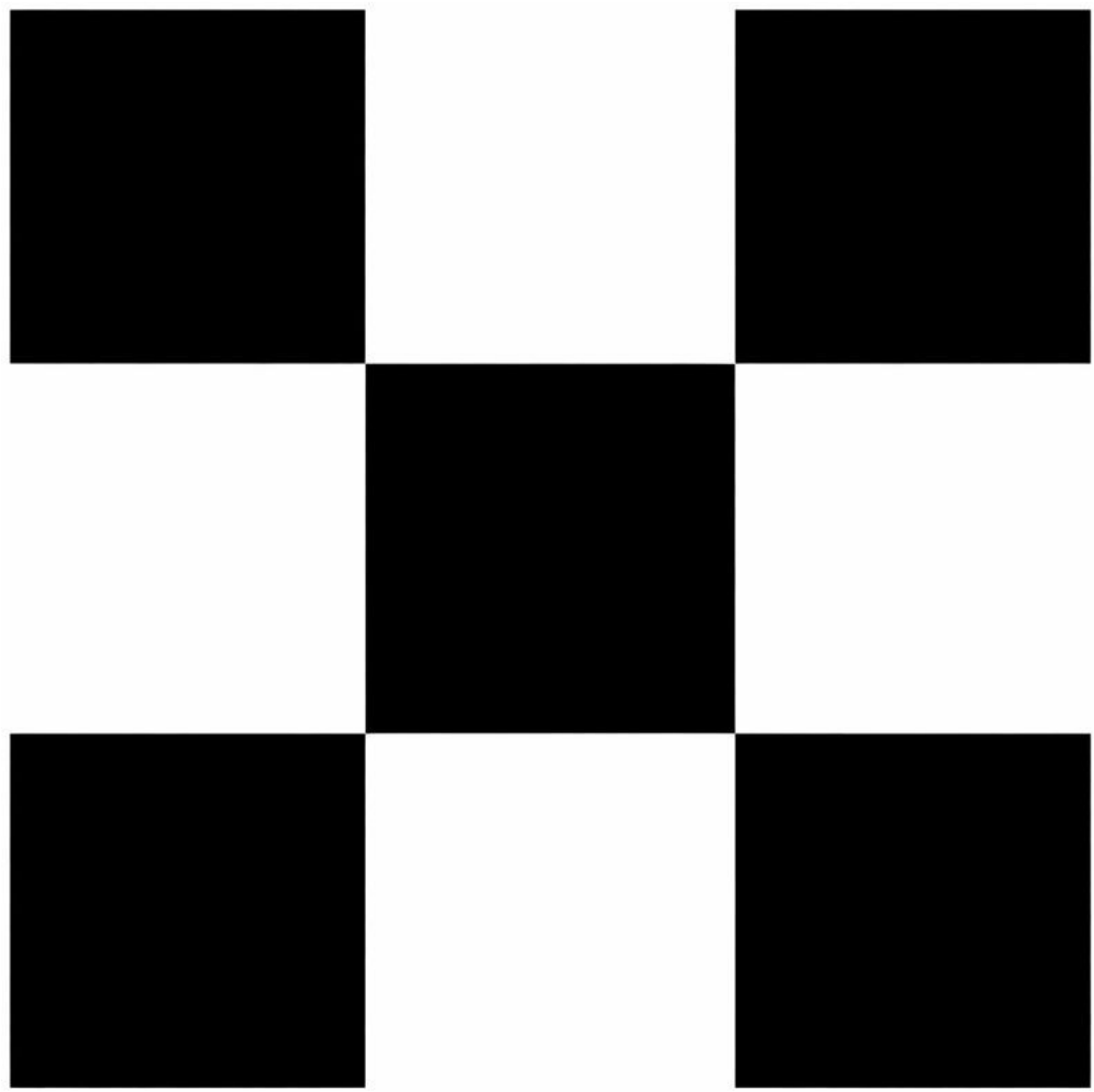
```
pip install opencv-python
```

```
Requirement already satisfied: opencv-python in c:\users\dell\
anaconda3\lib\site-packages (4.12.0.88)
Requirement already satisfied: numpy<2.3.0,>=2 in c:\users\dell\
anaconda3\lib\site-packages (from opencv-python) (2.2.6)
Note: you may need to restart the kernel to use updated packages.
```

```
pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\dell\anaconda3\
lib\site-packages (3.7.0)
Requirement already satisfied: numpy>=1.20 in c:\users\dell\anaconda3\
lib\site-packages (from matplotlib) (2.2.6)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (2.8.2)
Requirement already satisfied: packaging>=20.0 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (22.0)
Requirement already satisfied: pillow>=6.2.0 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (9.4.0)
Requirement already satisfied: cycler>=0.10 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (3.0.9)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (1.0.5)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\dell\
anaconda3\lib\site-packages (from matplotlib) (4.25.0)
Requirement already satisfied: six>=1.5 in c:\users\dell\anaconda3\
lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

```
# Displaying 18x18 pixel image.
Image(filename="C:/Users/DELL/Downloads/chessboard18x18.jpeg")
```



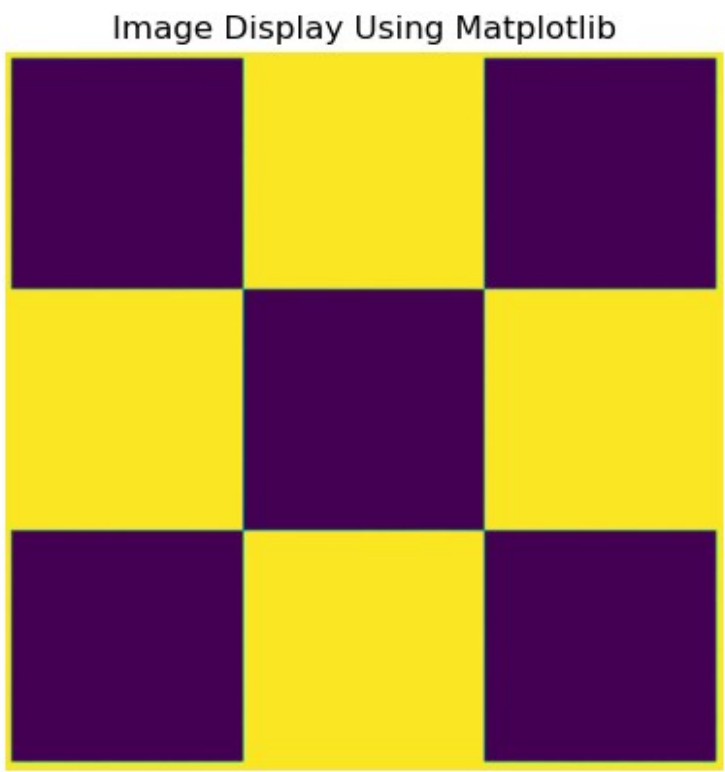
```
cb_img = cv2.imread("C:/Users/DELL/Downloads/chessboard18x18.jpeg",
                    cv2.IMREAD_GRAYSCALE)
print(cb_img)
```

```
[[253 249 253 ... 252 252 253]
 [255 255 255 ... 254 254 254]
 [253 255 255 ... 255 255 255]
 ...
 [254 254 254 ... 254 254 254]
 [254 254 254 ... 254 254 254]
 [254 254 254 ... 254 254 254]]
```

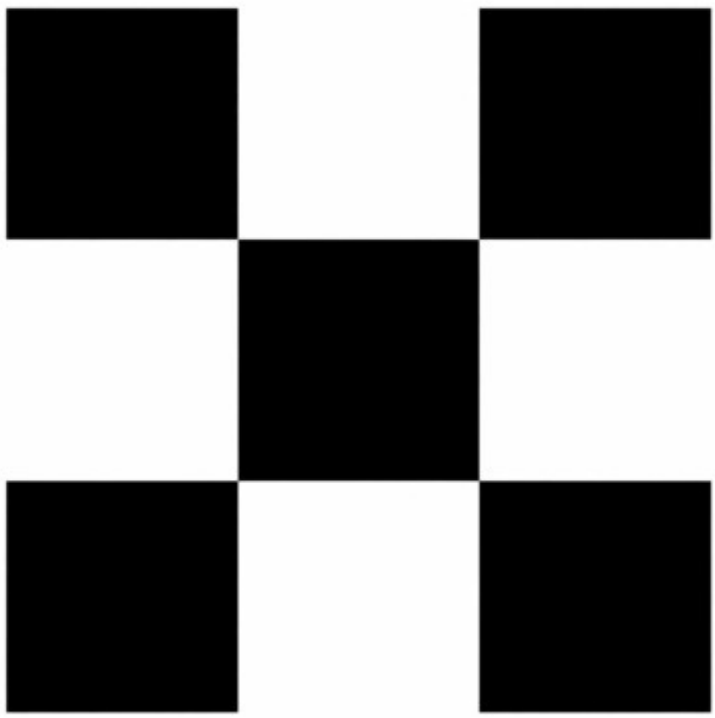
```python
# Displaying the image attributes
print("Shape of the image: ", cb_img.shape)
print("Data type of the image: ", cb_img.dtype)
print("Size of the image: ", cb_img.size)
```

```
Shape of the image:  (800, 800)
Data type of the image:  uint8
Size of the image:  640000
```

```python
# Displaying image using Matplotlib.
plt.imshow(cb_img)
plt.title("Image Display Using Matplotlib")
plt.axis("off") # To turn off axes
plt.show()
```



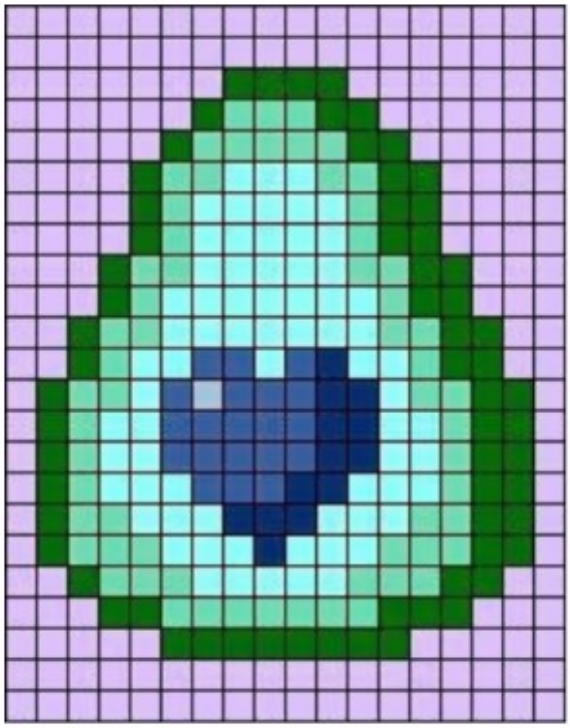

```python
# Set color map to gray scale for proper rendering.
plt.imshow(cb_img, cmap = "gray")
plt.axis("off")
plt.show()
```



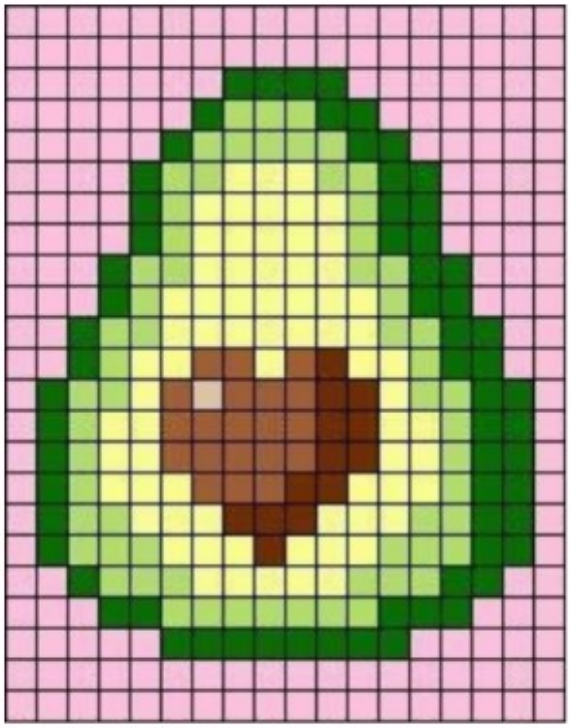
```
# Reading the image
avacado_img = cv2.imread("C:/Users/DELL/Downloads/avacado.png",
cv2.IMREAD_COLOR)

# Printing the shape of the image
print("Image shape (height, width, channels) is:", avacado_img.shape)

# Printing the size of the image
print("Image size is:", avacado_img.size)

# Printing the data-type of the image
print("Data type of image is:", avacado_img.dtype)
```

```
Image shape (height, width, channels) is: (478, 375, 3)
Image size is: 537750
Data type of image is: uint8
```

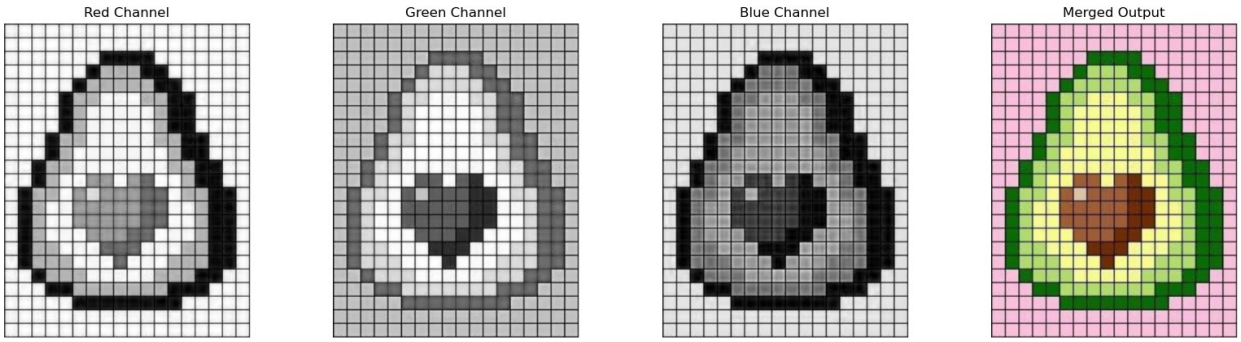
```
# Displaying the image using matplotlib
plt.imshow(avacado_img)
plt.axis("off")
plt.show()
```



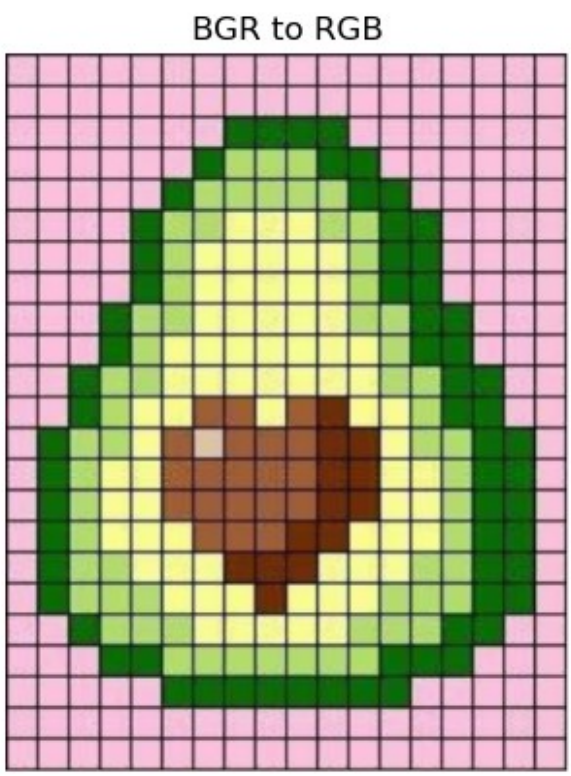
```
# Reversing the channels of the color image
avacado_img_channels_reversed = avacado_img[:, :, ::-1]
plt.imshow(avacado_img_channels_reversed)
plt.axis("off")
plt.show()
```



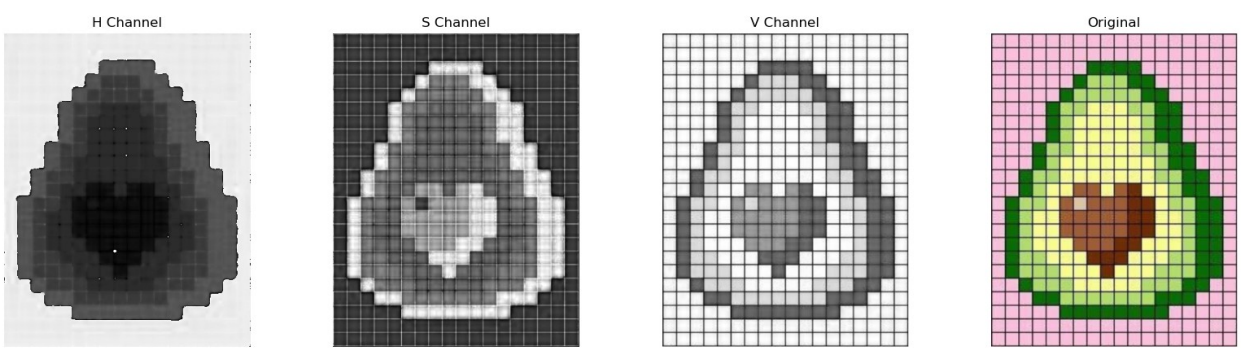
```
# Split the image into the B,G,R components
b, g, r = cv2.split(avacado_img)
# Show the channels
plt.figure(figsize = [20, 5])
# Red Channel
plt.subplot(1, 4, 1)
plt.imshow(r, cmap = "gray")
plt.title("Red Channel")
plt.axis("off")
# Green Channel
plt.subplot(1, 4, 2)
plt.imshow(g, cmap = "gray")
plt.title("Green Channel")
plt.axis("off")
# Blue Channel
plt.subplot(1, 4, 3)
plt.imshow(b, cmap = "gray")
plt.title("Blue Channel")
plt.axis("off")
# Merge the individual channels into a BGR image
merged_img = cv2.merge((b, g, r))
# Show the merged output
plt.subplot(1, 4, 4)
plt.imshow(merged_img[:, :, ::-1])
plt.title("Merged Output")
plt.axis("off")
plt.show()
```



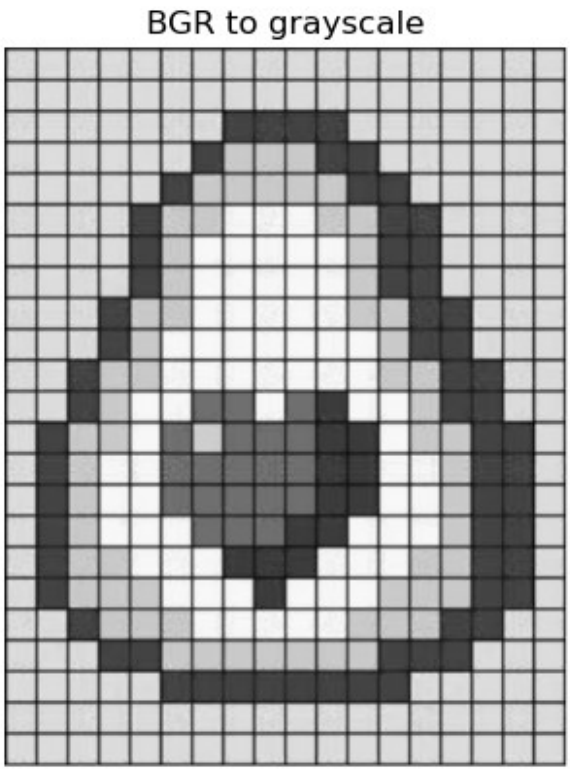

```
import cv2
import matplotlib.pyplot as plt

# OpenCV stores color channels in a differnet order than most other
applications
avacado_img_bgr = cv2.imread("C:/Users/DELL/Downloads/avacado.png",
cv2.IMREAD_COLOR)

# Converting BGR to RGB
avacado_img_rgb = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2RGB)

plt.imshow(avacado_img_rgb)
plt.title("BGR to RGB")
plt.axis("off")
plt.show()
```



```
# Converting to HSV
avacado_img_hsv = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2HSV)
# Split the image into H, S, V components
h, s, v = cv2.split(avacado_img_hsv)
# Show the channels
plt.figure(figsize = [20, 5])
# Hue Channel
plt.subplot(1, 4, 1)
plt.imshow(h, cmap = "gray")
plt.title("H Channel")
plt.axis("off")
# Saturation Channel
plt.subplot(1, 4, 2)
plt.imshow(s, cmap = "gray")
plt.title("S Channel")
plt.axis("off")
# Value Channel
plt.subplot(1, 4, 3)
plt.imshow(v, cmap = "gray")
plt.title("V Channel")
plt.axis("off")
# Show the original image
plt.subplot(1, 4, 4)
plt.imshow(avacado_img_rgb)
plt.title("Original")
plt.axis("off")
plt.show()
```

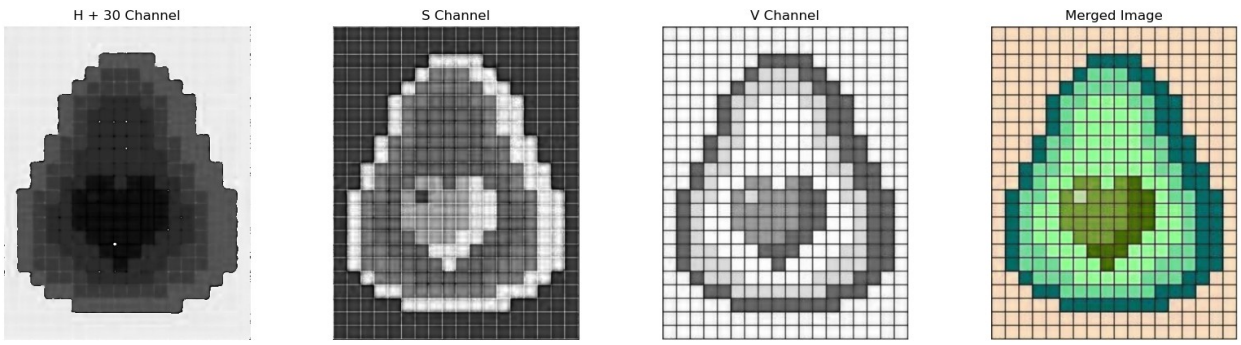



```
avacado_img_gray = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2GRAY)

plt.imshow(avacado_img_gray, cmap="gray")
plt.title("BGR to grayscale")
plt.axis("off")
plt.show()
```




```
h_new = h + 30
avacado_img_merged = cv2.merge((h_new, s, v))
avacado_img_rgb = cv2.cvtColor(avacado_img_merged, cv2.COLOR_HSV2RGB)

# Split the image into the B,G,R components
h,s,v = cv2.split(avacado_img_merged)

# Show the channels
plt.figure(figsize = [20, 5])

# Hue Channel
plt.subplot(1, 4, 1)
plt.imshow(h, cmap = "gray")
plt.title("H + 30 Channel")
plt.axis("off")

# Saturation Channel

plt.subplot(1, 4, 2)
plt.imshow(s, cmap = "gray")
plt.title("S Channel")
plt.axis("off")
# Value Channel
plt.subplot(1, 4, 3)
plt.imshow(v, cmap = "gray")
plt.title("V Channel")
```

```
plt.axis("off")
# Show the original image
plt.subplot(1, 4, 4)
plt.imshow(avacado_img_rgb)
plt.title("Merged Image")
plt.axis("off")
plt.show()
```




```
# Saving the image
cv2.imwrite("avacado_saved.jpg", avacado_img_rgb)
```

```
True
```